

Training Bias Trap

That was the first accuracy trap. The second one is different. Sometimes AI doesn’t invent anything. Every fact in the answer is real, but the picture is still distorted.

That’s training bias. The model learns whatever patterns are in the data it saw. If the data is uneven, incomplete, or full of shortcuts, the model can treat a narrow slice of reality as the default.

Imagine the only driving advice online came from professional race car drivers. If you asked, “What’s the best way to drive?” the model might tell you to drive 180 miles per hour on a closed track.

That answer wouldn’t come from evil intent. It would come from a narrow slice of data treated like the whole picture. **The trap is treating common as neutral.**

HOW BIAS GETS IN

Not one mechanism. Four overlapping ones, all rooted in the data the model saw.

**Overrepresentation**

Some cases appear too often in the data. The model treats them as the default.

**Underrepresentation**

Some cases appear too rarely. The model performs worse when those cases come up.

**Shortcuts**

A wrong clue happens to work in training. The model learns the clue instead of the real concept.

**Biased labels**

The labels come from past human decisions. If those decisions were biased, the model learns the bias.

Some of these are real cases. Some are simplified versions of a real pattern. In each one, the question is the same: what did the model actually learn?

FIGHT THE TRAP

Four moves when the answer feels too clean.

Spotting the trap is half the work. Pushing back is the other half. When an AI answer feels suspiciously default, here are the moves to try.

**Ask who’s missing.**

The AI gives you what’s most documented. Push it to surface what isn’t.

Try: “Whose perspective isn’t in this answer?”

**Ask for variation.**

The default is rarely the full range. Asking for variety forces the model past the top of the distribution.

Try: “Give me examples from different backgrounds, regions, and time periods.”

**Question the default.**

“Typical” often means “most common in the data,” which isn’t the same as “normal in the world.”

Try: “When you say “typical,” what assumptions are you making?”

**Slow down for decisions about people.**

Hiring, grading, discipline, medical, legal. AI is a starting point, not the final word. Humans review.

🔑 Common is not neutral. Frequent is not fair. Questioning the default is the skill.